

Tutorial 6: Higher Order Linear ODEs

[Solution]

1. Solve the homogeneous DE

$$y''' + 3y'' + 3y' + y = 0,$$

and find an appropriate trial solution (do not solve for the constants) of

$$y''' + 3y'' + 3y' + y = 2e^{-x} + \sin x \cos x.$$

Solution: Characteristic equation

$$r^3 + 3r^2 + 3r + 1 = 0 \Rightarrow r = -1 \quad (\text{multiplicity } 3)$$

3 linearly independent solutions are s:

$$y_1 = e^{-x}, \quad y_2 = xe^{-x}, \quad y_3 = x^2e^{-x}$$

Trial solution

$$y_{p1} = Ax^3e^{-x}, \quad y_{p2} = B \cos 2x + C \sin 2x.$$

So $y_p = Ax^3e^{-x} + B \cos 2x + C \sin 2x$.

2. Find the general solution to the following equations

(a) $y''' - y'' + y' - y = e^{-x} \sin x.$

Solution:

CE: $r^3 - r^2 + r - 1 = 0$, $r_1 = 1$, $r_2 = i$, $r_3 = -i$.

Trial solution, $y_p = (A \cos t + B \sin t)e^{-t}$ and we find that,

$$y_p''' - y_p'' + y_p' - y_p = e^{-x}(5A \cos x - 5B \sin x).$$

Comparing with RHS, we have $A = \frac{-1}{5}$ and $B = 0$.

So the general solution is

$$y(x) = \frac{-1}{5} \cos x e^{-x} + c_1 e^x + c_2 \cos x + c_3 \sin x$$

$$(b) \quad y''' + 3y'' + 3y' + y = \frac{2e^{-x}}{1+x^2}.$$

Solution:

$$\text{CE: } r^3 + 3r^2 + 3r + 1 = 0, \Rightarrow r = -1 \quad (\text{multiplicity } 3)$$

$$y_1 = e^{-x}, \quad y_2 = xe^{-x}, \quad y_3 = x^2e^{-x}$$

We use the method of variation of parameters. Consider $y_p = u_1y_1 + u_2y_2 + u_3y_3$ and $v_i = u'_i$,

$$\begin{cases} e^{-x}v_1 + xe^{-x}v_2 + x^2e^{-x}v_3 = 0 \\ (e^{-x})'v_1 + (xe^{-x})'v_2 + (x^2e^{-x})'v_3 = 0 \\ (e^{-x})''v_1 + (xe^{-x})''v_2 + (x^2e^{-x})''v_3 = \frac{2e^{-x}}{1+x^2} \end{cases}$$

So

$$\begin{cases} v_1 + xv_2 + x^2v_3 = 0 \\ -v_1 + (1-x)v_2 + (2x-x^2)v_3 = 0 \\ v_1 + (x-2)v_2 + (x^2-4x+2)v_3 = \frac{2}{1+x^2} \end{cases}$$

$$\begin{cases} v_1 + xv_2 + x^2v_3 = 0 \\ v_2 + 2xv_3 = 0 \\ v_3 = \frac{1}{1+x^2} \end{cases}$$

$$\begin{cases} u_3 = \int \frac{1}{1+x^2} dx = \tan^{-1} x \\ u_2 = \int (-2x) \frac{1}{1+x^2} = -\ln(1+x^2) \\ u_1 = x - \tan^{-1} x \end{cases}$$

$$y = e^{-x}((x^2 - 1)\tan^{-1} x + x - x \ln(1+x^2)) + (c_1 + c_2x + c_3x^2)e^{-x}$$

3. (a) Solve the equation

$$y''' + y' = \sec x.$$

Solution: We first solve $y''' + y' = 0$, the CE is:

$$r^3 + r = 0 \Leftrightarrow r(r^2 + 1) = 0 \Leftrightarrow r = 0, \pm i.$$

3 linearly independent solutions are:

$$y_1 = 1, y_2 = \cos x, y_3 = \sin x$$

To find one particular solution, we use the variation of parameter method.

We look for one particular solution of the form

$$y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

where u_1, u_2, u_3 satisfies,

$$\begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ u_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \sec x \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \\ u_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \sec x \end{pmatrix}$$

The augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & \cos x & \sin x & 0 \\ 0 & -\sin x & \cos x & 0 \\ 0 & -\cos x & -\sin x & \sec x \end{array} \right) \Leftrightarrow \left(\begin{array}{ccc|c} 1 & \cos x & \sin x & 0 \\ 0 & -\sin x & \cos x & 0 \\ 1 & 0 & 0 & \sec x \end{array} \right)$$

Hence

$$\begin{cases} u_1' = \sec x \\ u_2' = u_3' \frac{\cos x}{\sin x} = -1 \\ u_3' = -\frac{\sin x}{\cos x} \end{cases} \rightarrow \begin{cases} u_1 = \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| \\ u_2 = -x \\ u_3 = \ln |\cos x| \end{cases}$$

Therefore the general solution is:

$$y = \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| - x \cos x + \ln |\cos x| \sin x + c_1 + c_2 \cos x + c_3 \sin x.$$

(b)

(b) Solve the nonhomogeneous Euler's equation

$$x^3 y''' + x y' - y = 24x \ln x, \quad x > 0.$$

Solution:

Consider

$$\begin{aligned}
 x = e^t &\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \cdot \frac{dy}{dt} \\
 \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dt} \cdot \frac{dt}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \cdot \frac{dy}{dt} \right) = \frac{-dy}{x^2 dt} + \frac{1}{x} \frac{d^2y}{dt^2} \cdot \frac{1}{x} \\
 &= -\frac{dy}{x^2 dt} + \frac{1}{x^2} \frac{d^2y}{dt^2} = \frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \\
 \frac{d^3y}{dx^3} &= \frac{d}{dx} \left(\frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \right) \\
 &= -\frac{2}{x^3} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + \frac{1}{x^2} \left(\frac{d^3y}{dt^3} \cdot \frac{1}{x} - \frac{d^2y}{dt^2} \cdot \frac{1}{x} \right) \\
 &= \frac{1}{x^3} \frac{d^3y}{dt^3} - \frac{3}{x^3} \frac{d^2y}{dt^2} + \frac{2}{x^3} \frac{dy}{dt}
 \end{aligned}$$

Hence the original equation becomes:

$$\begin{aligned}
 &\left(\frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} \right) + \frac{dy}{dt} - y = 24e^t t \\
 \Leftrightarrow &y''' - 3y'' + 3y' - y = 24e^t t. \tag{1}
 \end{aligned}$$

We first solve $y''' - 3y'' + 3y' - y = 0$

$$y''' - 3y'' + 3y' - y = 0 \Leftrightarrow r = 1 \quad (\text{multiplicity } m = 3)$$

$\Rightarrow$ 3 linearly independent solutions:

$$y_1 = e^t, y_2 = te^t, y_3 = t^2e^t$$

To solve (1) we use undetermined coefficients method, the trial solution is:

$$y_p = t^3(At + B)e^t = (At^4 + Bt^3)e^t$$

Substituting into (1). We get

$$\begin{aligned}
 (24At + 6B - 18Bt)e^t &= 24e^t t \\
 \Rightarrow A = 1, B = 0 &\Rightarrow y_p = t^4e^t
 \end{aligned}$$

Hence the general solution:

$$y(t) = t^4 e^t + c_1 e^t + c_2 t e^t + c_3 t^2 e^t$$

By $t = \ln x$,

$$y(x) = x \ln^4 x + c_1 x + c_2 x \ln x + c_3 x \ln^2 x.$$

4. Consider the third-order equation

$$y''' + p(x)y'' + q(x)y' + r(x)y = f(x), \quad (2)$$

where p, q, r and f are continuous functions on some interval I containing 0.

- (i) Suppose that the equation (2) with $f(x) = 0$ and $p(x) = \tan x$ has solutions y_1, y_2 and y_3 , whose Wronskian at $x = 0$ is 1. Find the expression of their Wronskian.

Solution:

$$\text{Wronskian: } W(y_1, y_2, y_3)(x) = c \exp\left(\int_0^x p(t) dt\right)$$

$$\begin{aligned} W(y_1, y_2, y_3)(0) = 1 &\Rightarrow c = 1 \\ \Rightarrow W(y_1, y_2, y_3)(x) &= \exp\left(-\int_0^x \tan t dt\right) = \exp(t \ln |\cos x|) \\ &= |\cos x|. \end{aligned}$$

- (ii) Show that the equation (2) with $f(x) \neq 0$ always has four linearly independent solutions.

Solution:

Consider $y_p, y_p + y_1, y_p + y_2, y_p + y_3$, where y_p is a particular Solution to (1) & y_1, y_2, y_3 are 3 linearly independent solutions to

$$y''' + p(x)y'' + q(x)y' + r(x)y = 0.$$

Because y_p is a solution to (1), $y_p \notin \text{span}\{y_1, y_2, y_3\}$

Consider:

$$\exists c_i \quad s.t. \quad c_1 y_p + c_2 (y_p + y_1) + c_3 (y_p + y_2) + c_4 (y_p + y_3) = 0$$

$$\Leftrightarrow (c_1 + c_2 + c_3 + c_4) y_p + c_2 y_1 + c_3 y_2 + c_4 y_3 = 0$$

Because y_p, y_1, y_2, y_3 are linearly independent,

$$\begin{cases} c_1 + c_2 + c_3 + c_4 = 0 \\ c_2 = 0 \\ c_3 = 0 \\ c_4 = 0 \end{cases} \Leftrightarrow \begin{cases} c_1 = 0 \\ c_2 = 0 \\ c_3 = 0 \\ c_4 = 0 \end{cases}$$

Hence $y_p, y_p + y_1, y_p + y_2, y_p + y_3$ are linearly independent.